EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)

Topic	B: Complex Numbers
Sub topic	B5: Modulus, Argument, and Cartesian Forms

Mark Scheme

EXAM PAPERS PRACTICE

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic	**B: Complex Numbers**
Sub topic	**B5: Modulus, Argument, and Cartesian Forms**

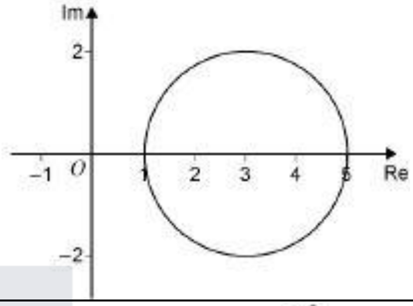
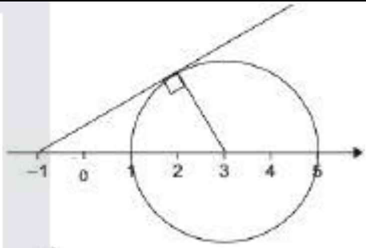
EXAM PAPERS PRACTICE
© 2024 Exams Papers Practice. All Rights Reserved

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

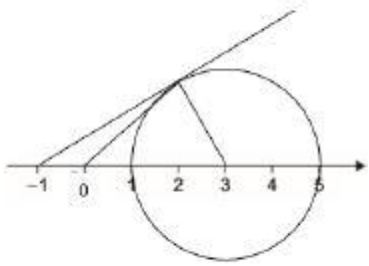
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Draws a circle with centre (3,0) and radius 2. Accept freehand circle. Ignore any straight lines drawn on the diagram.	AO1.1b	B1	
(b)(i)	Uses fully correct method for sin α or cos α or tan α $\sin\alpha = \frac{2}{4}$ $\cos\alpha = \frac{\sqrt{4^2-2^2}}{4}$ $\tan\alpha = \frac{2}{\sqrt{4^2-2^2}}$	AO3.1a	M1	
	Obtains correct value for α Accept 0.52(35987756) Condone 30°	AO1.1b	A1	$\sin\alpha = \frac{2}{4}$ $\alpha = \sin^{-1}\left(\frac{2}{4}\right)$ $\alpha = \frac{\pi}{6}$

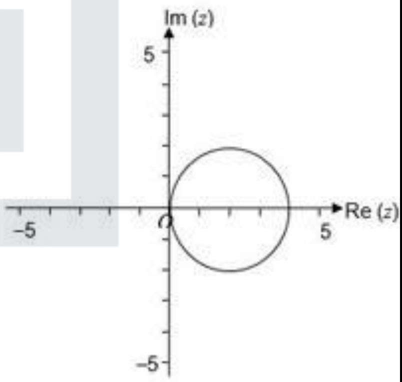
EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

	Marking Instructions	AO	Marks	Typical Solution
(ii)	Forms an equation in r using cosine rule or equivalent. Follow through their α. Or forms a correct equation in x and y.	AO3.1a	M1	 $r^2 = 2^2 + 3^2 - 2\times2\times3\times\cos\frac{\pi}{3}$ $r = \sqrt{7}$
	Forms an equation in θ using sine rule or equivalent. Follow through their α. Or forms a second correct equation in x and y.	AO1.1a	M1	$\frac{\sin\theta}{2} = \frac{\sin\frac{\pi}{3}}{\sqrt{7}}$
	Obtains correct value for r **or** θ.	AO1.1b	A1	$\sin\theta = 2\times\frac{\sqrt{3}}{2}\div\sqrt{7} = \frac{\sqrt{21}}{7}$

Or obtains correct values for x **and** y.			$\theta = 0.71$
Expresses w in the required form. Accept 2.6[45751311] or $\sqrt{7}$ for r Accept 0.71[37243789] or $\sin^{-1}\left(\frac{\sqrt{21}}{7}\right)$ or $\sin^{-1}\left(\sqrt{\frac{3}{7}}\right)$ for θ	AO1.1b	A1	$w = 2.6(\cos 0.71 + \mathrm{i} \sin 0.71)$
			Total 7 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Draws 'circle' with centre 2 + 0i Ignore other features	AO1.1a	M1	
	Draws circle passing through (0, 0), (4, 0), close to (2, 2) and (2, –2) with Imaginary axis tangential	AO1.1b	A1	
(b)	Uses mod/arg forms	AO3.1a	M1	$z - 2 = 2\left[\cos\left(-\frac{\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{\pi}{3}\right)\right]$
	Substitutes exact values for cos and sin Allow one slip	AO1.1a	M1	$= 2\left[\frac{1}{2} + \mathrm{i}\left(-\frac{\sqrt{3}}{2}\right)\right]$
	Obtains result in exact form	AO1.1b	A1	$z = 3 - \sqrt{3}\mathrm{i}$
				Total 5 marks

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q3.

Marking Instructions	AO	Marks	Typical Solution
Uses De Moivre's theorem	AO3.1a	M1	$(\cos\theta + \mathrm{i}\sin\theta)^5 = \frac{1}{\sqrt{2}}(1 - \mathrm{i})$
Equates real and imaginary parts and obtains two equations	AO1.1a	A1	$\Rightarrow \cos 5\theta + \mathrm{i}\sin 5\theta = \frac{1}{\sqrt{2}}(1 - \mathrm{i})$
Deduces that the smallest possible value of 5θ is	AO2.2a	A1F	

$\frac{7\pi}{4}$ FT from 'their' equations provided M1 has been awarded			$\cos 5\theta = \frac{1}{\sqrt{2}}$ $\sin 5\theta = -\frac{1}{\sqrt{2}}$ $(5\theta =)\frac{7\pi}{4}$ $\theta = \frac{7\pi}{20}$
Obtains the smallest possible value of θ from fully correct reasoning FT from 'their' 5θ provided M1 has been awarded	AO1.1b	A1F	
			Total 4 marks

Q4.

(a) $|4 - 4i| = \sqrt{16+16} = \sqrt{32} = 4\sqrt{2}$

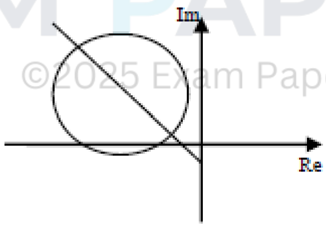
verification that $|-2+i+6-5i| = 4\sqrt{2}$

B1

$\arg(-2 + 2i) = \pi - \tan^{-1}(1) = \frac{3\pi}{4}$

verification that $\arg(z + i) = \frac{3\pi}{4}$

B1



2

(b) Circle

freehand circle sketched

M1

Centre at − 6 + 5i

clear from diagram or centre stated

A1

Cutting Re axis but not cutting Im axis

A1

"Straight" line

freehand line

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

M1

Half line from 0 − i

not horizontal or vertical but end point at 0 − i must be clear from diagram / stated

A1

gradient −1 (approx)

making 45° to negative Re axis and positive Im axis

A1

6

(c) Calculation based on fact that L_2 passes through centre of L_1

idea of vector $\begin{bmatrix} -4 \\ 4 \end{bmatrix}$ *from centre*

M1

Q represents −10 + 9i

must write as a complex number

A1

2

[10]

Q5.

(a) $r = 8$

B1

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

$\tan^{-1} \pm \frac{4\sqrt{3}}{4}$ or $\pm \frac{\pi}{3}$ seen

or $\frac{\pi}{6}$ *marked as angle to Im axis with "vector" in second quadrant on Arg diag*

M1

$\Rightarrow \theta = \frac{2\pi}{3}$

$-4 + 4\sqrt{3}\mathrm{i} = 8\,\mathrm{e}^{\mathrm{i}\frac{2\pi}{3}}$

A1

3

(b) (i) modulus of each root = 2

B1√

use of De Moivre − dividing argument by 3

M1

$\Rightarrow \theta = \frac{4\pi}{9}, \frac{2\pi}{9}, \frac{8\pi}{9}$

A1 if 3 "correct" values not all in requested interval

$2e^{-i\frac{4\pi}{9}}, \ 2e^{i\frac{2\pi}{9}}, \ 2e^{i\frac{8\pi}{9}}$

A2

4

(ii) Area = 3 × $\frac{1}{2}$ × PO × OR × sin $\frac{2\pi}{3}$

Correct expression for area of triangle PQR

M1

= 3 × $\frac{1}{2}$ × 2 × 2 × sin $\frac{2\pi}{3}$

correct values of lengths in formula

A1

$= 3\sqrt{3}$

A1cso

3

(c) Sum of roots (of cubic) = 0

must be stated explicitly

E1

Sum of 3 roots including Im terms

in form $r(\cos\theta + i\sin\theta)$

M1

$2\left(\cos\frac{(-)4\pi}{9} + \cos\frac{2\pi}{9} + \cos\frac{8\pi}{9}\right)$

isolating real terms; correct and with "2"

A1

$e^{-i\frac{4\pi}{9}} = \cos\frac{4\pi}{9} - i\sin\frac{4\pi}{9}$ seen earlier

or $\cos\frac{-4\pi}{9} = \cos\frac{4\pi}{9}$ *explicitly stated to earn final A1 mark*

$\cos\frac{2\pi}{9} + \cos\frac{4\pi}{9} + \cos\frac{8\pi}{9} = 0$

AG

A1**cso**

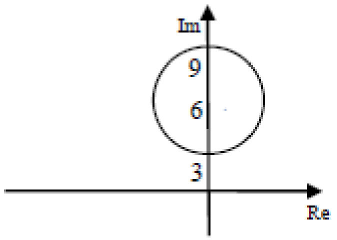
4

[14]

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q6.

(a)



Circle

freehand circle

M1

Centre at 6i

6 marked on Im axis as centre

A1

Radius 3 & cutting positive Im axis twice

radius of 3 clearly indicated with circle in position shown

A1

3

(b) (i) (Max |z| is) 9

1

B1

1

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(ii) Tangent from *O* to circle

FT their circle position

M1

Angle of $\frac{\pi}{6}$ or $\frac{\pi}{3}$ *correctly* marked

PI; condone degrees for first A1

A1

(Max arg *z* is) $\frac{2\pi}{3}$

exactly this

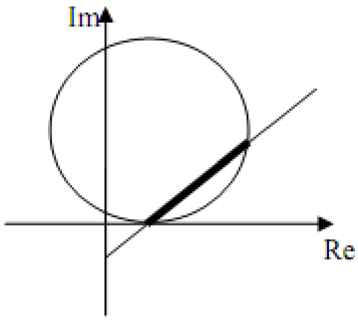
A1cso

3

[7]

Q7.

(a)



Use average of whole question if 2 diagrams used

(i) Circle

Circle in any position

B1

correct centre

Must be shown

B1

touching x–axis

ft incorrect centre

B1F

3

(ii) half–line

B1

through (0, –2)

Can be inferred

B1

through point of contact of circle with x–axis

B1

3

(b) Inside circle

B1

On line

ft errors in position of line and circle

B1F

2

[8]

Q8.

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(a) (i) $z = -\text{i}$ $\left|-2\sqrt{3} - 2\text{i}\right| = \sqrt{12+4} = 4$

$\left|-2\sqrt{3} - 2i\right|$

M1

4

A1

2

(ii) Centre of circle is $2\sqrt{3}$ + i

Do not accept $(2\sqrt{3},\ 1)$ *unless attempt to solve using trig*

B1

Substitute into line

M1

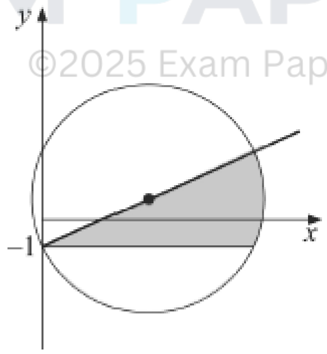
arg $(2\sqrt{3} + 2\text{i}) = \frac{\pi}{6}$ shown

A1

3

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(b)



Circle: centre correct

B1

through (0, –1)

B1

Half line: through (0, –1)

B1

through centre of circle

B1 4

(c) Shading inside circle and below line

B1F

Bounded by $y = -1$

B1 2

[11]

Q9.

(a) radius $\sqrt{2}$ centre $-5 + i$

condone (–5, 1) for centre do not accept (–5, i)

B1,B1 2

(b) arg (z_1 + 2i) = arg (–4 + 4i)

M1

$= \frac{3\pi}{4}$

clearly shown eg $\tan^{-1}\left(-\frac{1}{1}\right)$

A1 2

(c) (i) $|z_1 + 5 - i| = |1 + i| = \sqrt{2}$

B1 1

(ii) Gradient of line from

(–5, 1) to (–4, 2) is 1 $\left(\frac{\pi}{4}\right)$

M1 for a complete method

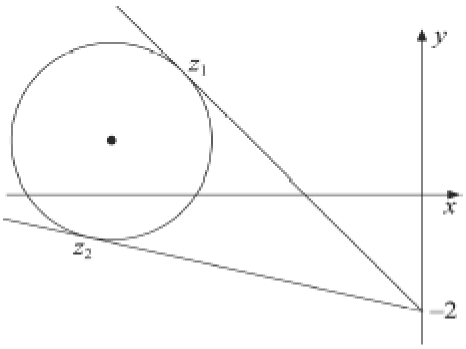
M1A1

radius $\perp$ line $\therefore$ tangent

E1 3

(iii)

©2025 Exam Papers Practice. All rights reserved.



Circle correct

ft incorrect centre or radius

B1F

Half line correct

line must touch C generally above the circle

B1

2

(d) z_2 in correct place

B0 if z_2 is directly below the centre of C

B1

with tangent shown

B1

2

[12]

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q10.

(a) $\frac{1+i}{1-i} = \frac{(1+i)^2}{1-i^2} = i$

AG

M1A1

2

(b) $|z_2| = \sqrt{\frac{1}{4}+\frac{3}{4}} = 1 = |z_1|$

M1A1

2

(c) $r = 1$

PI

B1

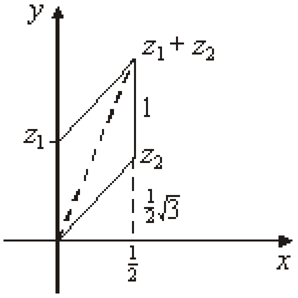
$\theta = \frac{1}{2}\pi, \frac{1}{3}\pi$

Deduct 1 mark if extra solutions

B1B1

3

(d)



Positions of the 3 points, relative to each other, must be approximately correct

B2,1F

2

(e) Arg(z_1+z_2) = $\frac{5}{12}\pi$

***Clearly** shown*

B1

$\tan\frac{5}{12}\pi = \dfrac{1+\frac{1}{2}\sqrt{3}}{\frac{1}{2}}$

Allow if B0 earned

M1

$= 2 + \sqrt{3}$

AG must earn B0 for this

B1

3

[12]

Q11.

(a) (i) $z + \frac{1}{z} = \cos\theta + \mathrm{i}\sin\theta + \cos(-\theta) + \mathrm{i}\sin(-\theta)$

Or $z + \frac{1}{z} = e^{i\theta} + e^{-i\theta}$

M1

$= 2\cos\theta$

AG

©2025 Exam Papers Practice. All rights reserved.

A1 **2**

(ii) $z^2 + \frac{1}{z^2}$ = cos 2θ + i sin 2θ + cos(-2θ) + i sin(-2θ)

M1

= 2 cos 2θ

OE

A1 **2**

(iii) $z^2 - z + 2 - \frac{1}{z} + \frac{1}{z^2}$

= 2cos 2θ − 2cosθ + 2

M1

Use of cos 2θ = 2 cos$^2\theta$ − 1

m1

= 4cos$^2\theta$ − 2cosθ

AG

A1 **3**

(b) $z + \frac{1}{z} = 0$ $z = \pm i$

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

M1A1

$z + \frac{1}{z} = 1$ $z^2 - z + 1 = 0$

M1A1

$z + \frac{1 \pm i\sqrt{3}}{2}$

A1F

Alternative:

cosθ = 0 $\theta = \pm \frac{1}{2}\pi$

M1

z = ±i

A1

cosθ = $\frac{1}{2}$ $\theta = \pm \frac{1}{3}\pi$

M1

z = $e^{\pm\frac{1}{3}\pi i} = \frac{1}{2}(1 \pm i\sqrt{3})$

A1A1

5

Accept solution to (b) if done otherwise

Alternative

If θ = + $\frac{1}{2}\pi$ θ = $\frac{1}{3}\pi$

M1

z = i $z = \frac{1+\sqrt{3}i}{2}$

A1

Or any correct z values of θ

M1

Any 2 correct answers

A1

One correct answer only

B1

[12]

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Examiner reports

Q1.

Part (a) was well answered, although most circles were drawn freehand. However, an accurately drawn circle made both (b)(i) and (b)(ii) easier to solve.

Most students realised that the required line was a tangent to the circle, but many of these failed to realise that a simple sine ratio could be used to find the angle α.

Part (b)(ii) was only answered well by only the most able students. Although most of the successful attempts used basic GCSE trigonometry, some elaborate methods were also employed, including the solution of quadratic simultaneous equations. A common error was to assume that the triangle with vertices 0, 3 and w was right-angled. Another common mistake was to assume that θ and α were equal.

Q4.

(a) Very few students scored full marks for the verification. It was not sufficient to merely write $|4 - 4\mathrm{i}| = 4\sqrt{2}$; some indication as to how the modulus was being found needed to be shown.

Similarly " $\arg(-2 + 2\mathrm{i}) = -\frac{\pi}{4} = \frac{3\pi}{4}$ ", which was presented as a solution by many, was not deemed worthy of a mark for verification. A sketch with a calculation often convinced examiners that the argument had the correct value.

(b) It was pleasing to see how well students understood these basic loci, and, although some could have taken greater care drawing a circle, a freehand sketch was acceptable. The main error was in drawing the half line where quite a few did not realise that it started at $0 - \mathrm{i}$ on the Argand diagram.

©2025 Exam Papers Practice. All rights reserved.

(c) Those who realised that the half line passed through the centre of the circle had little trouble in finding the complex number represented by Q. Another successful approach involved forming solving a quadratic equation based on the Cartesian equations of the circle and line. A few did the hard part of the question but wrote their final answer as coordinates and so only scored the method mark.

Q5.

(a) Many students seemed unable to find the correct argument of the complex number. Students would be well advised to make a small sketch of the Argand diagram in order to determine the correct quadrant for the angle θ .

(b) (i) An incorrect value of θ in part (a) prevented a large number of students from obtaining the correct roots of the cubic equation. Most students had the correct modulus and realised the need to divide an argument by 3, but did not always apply de Moivre's Theorem correctly.

(ii) It was possible for students to score full marks here if they had the correct modulus of the 3 roots, provided they recognised that the points P, Q and R were the vertices of an equilateral triangle. Poor trigonometrical skills prevented most students from obtaining the correct value for the area of the triangle.

(c) It was necessary to state explicitly that the sum of the roots of the cubic equation was zero. By considering the sum of the roots and then taking real parts of both sides of the equation the printed result followed. Better students stated explicitly that $\cos\left(-\frac{4\pi}{9}\right) = \cos\left(\frac{4\pi}{9}\right)$ at some stage in their proof.

Q6.

(a) Almost everyone managed to draw a circle for the locus with the vast majority realizing that the centre was at $6i$. Some drew a circle with its centre below the real axis and a few thought that the centre lay on the real axis. It was necessary to indicate that the radius of the circle was 3 and most did this by showing the values where the circle cut the imaginary axis. Others indicated the radius of 3 in a variety of ways but quite a few neglected to make this clear to examiners.

(b) (i) Several students wrote $9i$ instead of 9 as the maximum value of the modulus of z. It was evident that some had no idea what the question was really asking.

(ii) Most students realised the need to draw a tangent to the circle but weaker students could not cope with the basic trigonometry required to find a relevant angle on their diagram. Nevertheless a large number had no problem in finding the correct maximum value of the argument of z.

Q7.

The most satisfactory solutions came from candidates who were careful with their drawings and who used the same scale on both their axes. In part (a)(i), marks were generally lost by candidates either misplotting the centre of the circle, or by failing to realise that the circle touched the Real axis. In part (a)(ii), a number of candidates drew their half line from the origin and even when it did start at the point (0, –2), it failed to pass through the point of contact of the circle with the Real axis. Part (b) was not particularly well done as many candidates shaded a region within the circle or on occasions marked just the end points of the chord rather the full chord of the circle.

EXAM PAPERS PRACTICE ©2025 Exam Papers Practice. All rights reserved.

Q8.

Lack of clear evidence that candidates understood what they were doing in part (a) caused a loss of marks for this part of the question. Methods varied. Those candidates who turned this part of the question into a coordinate geometry exercise probably provided the clearest solutions.

Those candidates who evaluated $\left|-2i-2\sqrt{3}\right|$ and arg($2\sqrt{3}$ + 2i) provided less convincing solutions and in some cases evident error. For instance it was not uncommon to see $\left|-2i-2\sqrt{3}\right|$ written as $\sqrt{(2\sqrt{3})^2-(2i)^2}$. Part (b) on the whole was done well except that in some instances not all the results of part (a) were incorporated in the candidate's Argand diagram. Part (c) was done well but, if a mistake did occur, it was almost always that the shaded area would be bounded by the real axis rather than by a line parallel to the real axis through the point represented by the complex number z = –i.

Q9.

Apart from the odd sign error, part (a) was done correctly. Answers to part (b) and to some extent part (c)(i) lacked detail especially as the results were given or implied. Very few candidates scored credit in part (c)(ii) and when it was attempted it was almost always solved by finding the cartesian equation of the circle C and the line L and by showing that the resulting quadratic equation giving the points of intersection of C and L had two coincident roots. This method, of course, involved a considerable amount of algebraic manipulation, when in fact the consideration of the gradient of the line joining the centre of C to the point represented by z_1 would have given the result immediately. However, this was very rarely seen. It was surprising to see many candidates draw the circle C and the line L intersecting in two points on the Argand diagram (part (c)(iii)) in spite of the results candidates had been requested to show in the earlier parts of the question. There were many incorrect solutions to part (d), the most common being to place z_2 vertically below the centre of C.

Q10.

Part (a) was well done, as was part (b). In part (c) it was surprising to note how many candidates could not express the complex number in the form $re^{i\theta}$, although z_2 was almost invariably correctly written as $e^{i\frac{\pi}{3}}$ Errors in part (c) however did not deter candidates from drawing a correct Argand diagram as they usually used the form $\boldsymbol{a} + i\boldsymbol{b}$ when plotting their points. Although the vast majority of scripts ended with $\tan \frac{5\pi}{12} = 2 + \sqrt{3}$, very, very few candidates gave convincing proof that $\arg(z_1+z_2)$ was $\frac{5\pi}{12}$, but rather seemed to take it for granted.

Q11.

Parts (a) was quite well done and many candidates scored the available seven marks. There were however some serious algebraic errors, the commonest of which was to equate $z^2 + \frac{1}{z^2}$ to $\left(z + \frac{1}{z}\right)^2$ in part (a)(ii) with some consequent faking to arrive at the printed result in part (a)(iii). Part (b) however was very poorly attempted. Candidates did not seem to realise that z could be equal to zero and consequently multiplied $(\cos\theta + i\sin\theta)^2$ by $4\cos^2\theta - 2\cos\theta$ Of those candidates who realised that $4\cos^2\theta - 2\cos\theta$ was equal to zero, the factorisation of this quadratic in $\cos\theta$ evaded most and, even when attempts were made to solve $4\cos^2\theta - 2\cos\theta = 0$, the factor $\cos\theta$ disappeared and the other solution $\cos\theta = 1/2$ usually produced just one root from $\theta = \frac{\pi}{3}$. Candidates who were able to obtain both $\cos\theta = 0$ and $\cos\theta = 1/2$ usually produced only two solutions and subsequently two roots. It did not seem to occur to candidates that a quartic equation would have four roots.

©2025 Exam Papers Practice. All rights reserved